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Semi-supervised tracking in medical images with cycle tracking

Abstract

Systems and methods for tracking a location of an object of interest through a sequence of medical images are provided. First and second input medical images of a patient are received. The first input medical image comprises an annotation of a location of an object of interest. Features are extracted from the first and the second input medical images. A location of the object of interest in the second input medical image is determined using a machine learning based location predictor network based on the annotation of the location of the object of interest in the first input medical image and the extracted features from the first and the second input medical images. The location of the object of interest in the second input medical image is output. The machine learning based location predictor network is trained based on a comparison between 1) locations of a particular object in a sequence of training images determined during a forward tracking of the particular object through the sequence of training images and 2) locations of the particular object determined during a backward tracking of the particular object through the sequence of training images.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates generally to tracking in medical images, and in particular to semi-supervised tracking in medical images with cycle tracking.

BACKGROUND

(2) The tracking of objects of interest in medical images is an important task for many different clinical applications. One important clinical application of tracking objects of interest in medical images is visual guidance during interventional procedures. As an example, in the case of x-ray imaging, a contrast agent is administered to a patient for navigating in the patient. However, contrast agents can be harmful to the patient, cause longer procedure times, and cause prolonged exposure to radiation for both patients and clinicians. Further, the contrast agent may occlude portions of the x-ray images. While various methods have been proposed to address this occlusion, such methods present several constraints on their usage in clinical settings. For example, such methods may only work on images where no contrast agents are used or where human intervention is required for images acquired using contrast agents, may require assumptions on the shape of the devices tracked which may not be valid, or may have sub-optimal accuracy for clinical applications.

(3) Recently, deep learning based approaches for tracking objects of interest in images have been proposed. Such deep learning based approaches are trained on extensively annotated datasets of natural images. However, the application of such deep learning based approaches to medical images is challenging due to the difficulty, the time, and the expense in annotating medical images and due to the performance requirements in medical applications.

BRIEF SUMMARY OF THE INVENTION

(4) In accordance with one or more embodiments, systems and methods for tracking a location of an object of interest through a sequence of medical images are provided. First and second input medical images of a patient are received. The first input medical image comprises an annotation of a location of an object of interest. Features are extracted from the first and the second input medical images. A location of the object of interest in the second input medical image is determined using a machine learning based location predictor network based on the annotation of the location of the object of interest in the first input medical image and the extracted features from the first and the second input medical images. The location of the object of interest in the second input medical image is the output of the predictor network. The machine learning based location predictor network is trained based on a comparison between 1) locations of a particular object in a sequence of training images determined during a forward tracking of the particular object through the sequence of training images and 2) locations of the particular object determined during a backward tracking of the particular object through the sequence of training images.

(5) In one embodiment, the machine learning based location predictor network determines the locations of the particular object from a first image to a last image of the sequence of training images during the forward tracking and the machine learning based location predictor network determines the locations of the particular object from the last image to the first image in the sequence of training images during the backward tracking. A first image and a last image in the sequence of training images may be annotated with a ground truth location of the particular object.

The location of the object of interest in the first input medical image may be manually annotated by a user or automatically annotated using a machine learning based network

(6) In one embodiment, the machine learning based location predictor network may be further trained based on a comparison between locations of the particular object in annotated training images of the sequence of training images and annotated ground truth locations of the particular object in the annotated training images. In another embodiment, the machine learning based location predictor network may be further trained based on a comparison between a location of the particular object in a first training image of the sequence of training images and an annotated ground truth location of the particular object in the first training image of the sequence of training images.

(7) In one embodiment, the location of the object of interest in the second input medical image may be determined by correlating the features extracted from the first input medical image with the features extracted from the second input medical image and determining the location of the object of interest in the second input medical image based on the correlated features. The features may be correlated by matching features extracted from the first input medical image for regions corresponding to the location of the object of interest with features extracted from the second input medical image.

(8) In one embodiment, the receiving, the extracting, the determining, and the outputting are repeated using the second input medical image as the first input medical image and a next image of the sequence of medical images as the second input medical image.

(9) These and other advantages of the invention will be apparent to those of ordinary skill in the art by reference to the following detailed description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a method for tracking a location of an object of interest through in a sequence of input medical images, in accordance with one or more embodiments;
- (2) FIG. 2 shows a workflow for tracking a location of an object of interest through in a sequence of input medical images, in accordance with one or more embodiments;
- (3) FIG. 3 shows a workflow for training a machine learning based location predictor network for tracking a location of a particular object through a sequence of medical images, in accordance with one or more embodiments;
- (4) FIG. 4 shows augmented images for training a machine learning based location predictor network, in accordance with one or more embodiments;
- (5) FIG. 5 shows an exemplary artificial neural network that may be used to implement one or more embodiments;
- (6) FIG. 6 shows a convolutional neural network that may be used to implement one or more embodiments; and
- (7) FIG. 7 shows a high-level block diagram of a computer that may be used to implement one or more embodiments.

DETAILED DESCRIPTION

(8) The present invention generally relates to methods and systems for semi-supervised tracking in medical images with cycle tracking. Embodiments of the present invention are described herein to give a visual understanding of such methods and systems. A digital image is often composed of digital representations of one or more objects (or shapes). The digital representation of an object is often described herein in terms of identifying and manipulating the objects. Such manipulations are virtual manipulations accomplished in the memory or other circuitry/hardware of a computer system. Accordingly, is to be understood that embodiments of the present invention may be

performed within a computer system using data stored within the computer system.

(9) Embodiments described herein provide for the semi-supervised tracking of objects of interest through a sequence of input medical images using a machine learning based location predictor network. The location predictor network is trained, at least in part, with trajectory consistency. In trajectory consistency, the location predictor network is trained to first track a location of an object of interest forward through a sequence of training images and then backward through the sequence of training images. The location predictor network is trained with a trajectory consistency loss that compares locations of the object of interest determined in the forward tracking with locations of the object of interest determined in the backward tracking. Advantageously, the location predictor network is trained with trajectory consistency by comparing predicted locations with each other, and does not require ground truth locations of the object of interest annotated in each of the images in the sequence of training images.

(10) FIG. 1 shows a method **100** for tracking a location of an object of interest through in a sequence of input medical images, in accordance with one or more embodiments. The steps of method **100** may be performed by one or more suitable computing devices, such as, e.g., computer **702** of FIG. 7. FIG. 2 shows a workflow **200** for tracking a location of an object of interest through in a sequence of input medical images, in accordance with one or more embodiments. FIG. 1 and FIG. 2 will be described together.

(11) At step **102** of FIG. 1, first and second input medical images of a patient are received. The first input medical image comprises an annotation of a location of an object of interest. The object of interest may be any suitable object of interest, such as, e.g., a medical instrument (e.g., catheter), an anatomical landmark, a lesion, etc. The location of the object of interest may be manually annotated in the first input medical image by a user (e.g., a clinician) or may be automatically annotated, e.g., by an upstream system in the clinical workflow. For example, the location of the object of interest in the first input medical image may be automatically annotated by a machine learning based network determining a location of the object of interest in the first input medical image. In one embodiment, the location of the object of interest in the first input medical image may be determined during a prior iteration of method **100** of FIG. 1.

(12) In one embodiment, the first and second input medical images are patches extracted from medical images. For example, as shown in workflow **200** of FIG. 2, the first input medical image may be template patch **l.sub.t 204** extracted (e.g., cropped) from template image or frame **202** and the second input medical image may be search patch **l.sub.t+1 208** extracted from search image or frame **206**. Template patch **l.sub.t 204** may be extracted from template frame **202** centered around the location of the object of interest in the image **202**. Search patch **l.sub.t+1 208** may be extracted from search frame **206** centered around a location in search frame **206** corresponding to the location of the object of interest in template frame **202**. Template patch **l.sub.t 204** and search patch **l.sub.t+1 208** may be of any suitable predetermined size. In one embodiment, search patch **l.sub.t+1 208** is of a size larger than template patch **l.sub.t 204** to prevent situations where the object of interest moves outside of the cropped area due to a large motion. Template frame **202** is denoted as the “template” frame as template frame **202** comprises the annotation of the location of the object of interest while search frame **206** is denoted as the “search” frame as the location of the object of interest is to be determined in search frame **206** to thereby track the location of the object of interest from template frame **202** to search frame **206**. Advantageously, by extracting patches **204** and **208**, the input size of the input medical images is reduced to thereby enable near real-time performance. In another embodiment, the first and second input medical images are template frame **202** and search frame **206** respectively.

(13) In one embodiment, the first and second input medical images are consecutive images or frames of a sequence of input medical images. For example, as shown in workflow **200** of FIG. 2, template patch **204** is extracted from template frame **202** acquired at a time $T=t$ and search patch **208** is extracted from search from **206** acquired at a next time $T=t+1$.

(14) The first and second input medical images may be of any suitable modality, such as, e.g., CT (computed tomography), MRI (magnetic resonance imaging), ultrasound, x-ray, or any other medical imaging modality or combinations of medical imaging modalities. The first input medical image and/or the second input medical image may be 2D (two dimensional) images and/or 3D (three dimensional) volumes. The first input medical image and/or the second input medical image may be received directly from an image acquisition device, such as, e.g., a CT scanner, as the medical images are acquired, or can be received by loading previously acquired medical images from a storage or memory of a computer system or receiving medical images that have been transmitted from a remote computer system.

(15) At step **104** of FIG. **1**, features are extracted from the first and the second input medical images. The features may be extracted from the first and the second input medical images using any suitable approach. In one embodiment, the features may be extracted from the first and the second input medical images using a machine learning based feature extractor network, such as, e.g., a fully convolutional neural network. For example, as shown in workflow **200** of FIG. **2**, features **212** and features **214** are respectively extracted from template patch **204** and search patch **208** using feature extraction network **210**. While feature extraction network **210** is separately shown in workflow **200**, it should be understood that feature extraction network **210** for extracting features from template patch **204** and search patch **208** is the same feature extraction network. The features are low-level latent features representing the most important features from template patch **204** and search patch **208**. In one embodiment, the features may be represented in multiple channels each representing aspects of the images (e.g., shape, color, etc.).

(16) At step **106** of FIG. **1**, a location of the object of interest in the second input medical image is determined using a machine learning based location predictor network based on the annotation of the location of the object of interest in the first input medical image and the extracted features from the first and the second input medical images.

(17) In one embodiment, the location of the object of interest in the second input medical image is determined by first correlating the features extracted from the first input medical image with the features extracted from the second input medical image. The correlation may be performed using any suitable machine learning based correlation network. For example, as shown in workflow **200** of FIG. **2**, features **212** and **214** are correlated by depth wise cross correlation layer **216**. Depth wise cross correlation layer **216** matches features extracted from the first input medical image for regions corresponding to the location of the object of interest with features with features extracted from the second input medical image. For example, depth wise cross correlation layer **216** may compare a first channel of features extracted from template patch **204** with a corresponding first channel of features extracted from search patch **208**, a second channel of features extracted from template patch **204** with a corresponding second channel of features extracted from search patch **208**, etc. In this manner, the correlation of the features is performed in a three dimensionally across each channel of the features.

(18) The location predictor network then determines a location of the object of interest in the second input medical image based on the correlated features. The location of the object of interest in the second input medical image may be determined by any suitable machine learning based location predictor network, such as, e.g., a fully convolutional neural network. For example, as shown in workflow **200** of FIG. **2**, location **220** of the object of interest in template frame **202** is determined based on the correlated features by location predictor network **218**.

(19) The location predictor network is trained during a prior offline or training stage using one or more sequences of training images. During training, the location predictor network tracks the location of a particular object by performing a pairwise analysis of pairs of images in the sequence of training images during a forward tracking and a backward tracking. In the forward tracking, the location predictor network tracks the location from a first image in the sequence of training images to a last image in the sequence of training images. In the backward tracking, the location predictor

network tracks the location from the last image in the sequence of training images to the first image in the sequence of training images. While the first and the last images in the sequence of training images are annotated with the location of the particular object, the training images therebetween are not necessarily annotated. In one embodiment, the location predictor network is trained, at least in part, with a trajectory consistency loss based on a comparison between 1) locations of the particular object in the sequence of training images determined during the forward tracking of the particular object through the sequence of training images and 2) locations of the particular object determined during the backward tracking of the particular object through the sequence of training images. The location predictor network may additionally be trained based on a keyframe loss and/or a cycle loss. Training of the location predictor network is described in further detail below with respect to FIG. 3. Once trained, the location predictor network is applied during an online or inference stage, for example, to perform step **106** of FIG. 1.

(20) The location of the object of interest in the second input medical image may be represented in any suitable form. In one embodiment, the location of the object of interest in the second input medical image is a location relative to the location of the object of interest in the first input medical image. For example, the location of the object of interest in the second input medical image may be represented as offsets in the x-axis, the y-axis, and/or the z-axis) relative to a coordinate of the location of the object of interest in the first input medical image. In another embodiment, the location of the object of interest in the second input medical image is represented as a coordinate in the second input medical image.

(21) At step **108** of FIG. 1, location of the object of interest in the second input medical image is output. For example, the location of the object of interest in the second input medical image can be output by displaying the location of the object of interest in the second input medical image on a display device of a computer system, storing the location of the object of interest in the second input medical image on a memory or storage of a computer system, or by transmitting the location of the object of interest in the second input medical image to a remote computer system.

(22) In one embodiment, method **100** of FIG. 1 may be iteratively repeated any number of iterations to track the location of the object of interest through the sequence of medical images. For example, method **100** may be iteratively repeated using the second input medical image annotated with the determined location of the object of interest as the first input medical image and the next consecutive image in the sequence of medical images as the second input medical image.

(23) FIG. 3 shows a workflow **300** for training a machine learning based location predictor network for tracking a location of a particular object through a sequence of medical images, in accordance with one or more embodiments. Workflow **300** is performed during an offline or training stage. In one example, workflow **300** is performed to train the location predictor network applied at step **106** of FIG. 1. $t=1$

(24) In workflow **300**, the location predictor network is trained with a sequence of training images comprising training images **302**, **304**, **306**, and **308** respectively acquired at times $t=0$, $t=1$, $t=2$, and $t=3$. While the sequence of training images is illustratively shown in FIG. 3 as comprising four training images, the sequence of training images may comprise any suitable number of training images. The first training image **302** and the last training image **308** in the sequence of training images are annotated with the location of the particular object. The location predictor network is applied on pairs of training images in the sequence of training images to track a location of the particular object during a forward tracking and a backward tracking. During the forward tracking, the location predictor network is applied on pairs of training images starting from the first training image **302** to the last training image **308** to track the location of the particular object forward through the sequence of training images. For example, during the forward tracking, the location predictor network is applied to training images **302** and **304** to determine a position of the particular object in training image **304**, to training images **304** and **306** to determine a position of the particular object in training image **306**, and to training images **306** and **308** to determine a

position of the particular object in training image **308**. During the backward tracking, the location predictor network is applied on pairs of training images starting from the last training image **308** to the first training image **302** to track the location of the particular object backward through the sequence of training images. For example, during the backward tracking, the location predictor network is applied to training images **308** and **306** to determine a position of the particular object in training image **306**, to training images **306** and **304** to determine a position of the particular object in training image **304**, and to training images **304** and **302** to determine a position of the particular object in training image **302**. The operation of the location predictor network for determining the position of the particular object is similar to that described with respect to FIGS. **1** and **2**. The location predictor network is trained using a number of losses based on the forward and backward tracking. In one embodiment, the location predictor network is trained based on a trajectory consistency loss. In other embodiments, the location predictor network may additionally or alternatively be trained based on a keyframe loss and/or a cycle loss.

(25) In trajectory consistency, when the location predictor network tracks the particular object forward and backward through the sequence of training images, the predicted locations of the particular object should follow the same trajectory. According, the location predictor network is trained with a trajectory consistency loss to optimize (e.g., minimize) an error comparing the predicted location of the particular object in a training image *i* (e.g., training image **304** or **306**) during the forward tracking and the predicted location of the particular object in training image *i* during the backward tracking. Given a sequence of training images of length *N*, the trajectory consistency loss $L_{\text{sub.ccy}}$ may be defined as follows:

$$(26) \quad L_{\text{ccy}} = \frac{1}{N-1} \cdot \text{Math.} \left(L_{\text{forward}}^i - L_{\text{backward}}^i \right)^2 \quad \text{Equation(1)}$$

where $L_{\text{sub.forward.sup.i}}$ denotes the location of the particular object in training image *i* determined by the location predictor network during the forward tracking and $L_{\text{sub.backward.sup.i}}$ denotes the location of the particular object in training image *i* determined by the location predictor network during the backward tracking.

(27) In the keyframe loss, when the location predictor network tracks the particular object forward and backward through the sequence of training images, for training images annotated with ground truth locations of the particular object, the predicted locations of the particular object should be the same as the annotated ground truth locations of the particular object. Accordingly, the location predictor network is trained with keyframe loss to optimize (e.g., minimize) an error (e.g., a root mean square error) comparing the predicted location of the particular object in an annotated training image *i* (e.g., training image **302** or **308**) during the forward tracking or the backward tracking and the annotated ground truth location of the particular object in the annotated training image *i*. Given S_i annotated images, the keyframe loss $L_{\text{sub.kf}}$ may be defined as follows:

$$(28) \quad L_{\text{kf}} = \sqrt{\frac{1}{S_i} \cdot \text{Math.} \left(L_{\text{pred}}^i - L_{\text{GT}}^i \right)^2} \quad \text{Equation(2)}$$

where $L_{\text{sub.pred.sup.i}}$ denotes the location of the particular object in training image *i* determined by the location predictor network (during the forward tracking or the backward tracking) and $L_{\text{sub.GT.sup.i}}$ denotes the annotated ground truth location of the particular object in training image *i*.

(29) In cycle consistency, when the location predictor network tracks the particular object forward and backward through the sequence of training images, the predicted location of the particular object tracked forward and backward through the sequence of training images should end where it began. Accordingly, the location predictor network is trained with cycle consistency loss to optimize (e.g., minimize) an error comparing the predicted location of the particular object in the first training image $i=0$ in the sequence of training images and the annotated ground truth location of the particular object in the first training image $i=0$. The cycle consistency loss may be defined as follows:

$$L_{\text{sub.cycle}} = (L_{\text{sub.pred.sup.0}} - L_{\text{sub.GT.sup.0}})_{\text{sup.2}} \quad \text{Equation (3)}$$

where $L_{\text{sub.pred.sup.0}}$ denotes the location of the particular object in the first image $i=0$ in the sequence of training images determined by the location predictor network, with the location $L_{\text{sub.pred.sup.0}}$ obtained after forward tracking from the first frame to the last frame, and backward tracking from the location on the last frame back to the first frame, and $L_{\text{sub.GT.sup.0}}$ denotes the annotated ground truth location of the particular object in the first training image $i=0$. (30) The location predictor network may be trained with trajectory consistency loss, keyframe loss, and cycle consistency loss according to the following final loss function:

$$L_{\text{sub.total}} = \omega_{\text{sub.1}} L_{\text{sub.kf}} + \omega_{\text{sub.2}} L_{\text{sub.cycle}} + \omega_{\text{sub.3}} L_{\text{sub.ccy}} \quad \text{Equation (4)}$$

where $\omega_{\text{sub.1}}$, $\omega_{\text{sub.2}}$, and $\omega_{\text{sub.3}}$ are weights for the keyframe loss, the cycle consistency loss, and the trajectory consistency loss respectively. In one embodiment, the weights are defined as follows: $\omega_{\text{sub.1}}=1$, $\omega_{\text{sub.2}}=0.1$, and $\omega_{\text{sub.3}}=0.1$ such that the keyframe loss is assigned the highest weight. This is because, in the keyframe loss, the predicted locations are compared with annotated ground truth locations.

(31) In one embodiment, the location predictor network is trained with curriculum learning. In curriculum learning, the location predictor network is trained with relatively shorter sequences of training images in the beginning of training. The length of the sequences of training images increases (e.g., linearly) as the training progresses. The reason for this is because starting the training with relatively longer sequences of training images results in unstable training, as the initial values of the keyframe loss will be very high since the location predictor network is not yet efficient as predicting the location of the particular object. Curriculum learning ensures that the location predictor network is first able to manage relatively shorter sequences of training images. By increasing the length of the sequences of training images progressively, the data distribution does not change too quickly and the training can be stabilized.

(32) In one embodiment, the sequences of training images may be augmented to make feature extraction more robust to noise in images and to various degrees of image quality. FIG. 4 shows augmented images for training a machine learning based location predictor network, in accordance with one or more embodiments. Image 402 is an original image and images 404-410 are augmented images generated from image 402. Augmented image 404 is generated by applying noise following a Poisson distribution to original image 402. Poisson noise is present in x-ray images due to random processes associated with photons when the image is acquired. Augmented image 406 is generated by applying noise following a Gaussian distribution to original image 402. Augmented image 408 is generated by adding salt and pepper noise by randomly assigning an intensity value of 0 (i.e., black) or 1 (i.e., white) to pixels of original image 402. Augmented image 410 is generated by adding color jittering by randomly modifying contrast, brightness, hue, saturation values, and other image parameters of original image 402. While the augmentations described herein target the x-ray domain, any type of augmentation may be applied according to the target application.

(33) Advantageously, embodiments described herein track the locations of an object of interest though a sequence of input medical images while not making assumptions of the object of interest being tracked. The location predictor network does not require extensively annotated training images and may be trained with semi and self-supervised supervision signals. Embodiments described herein were experimentally validated and was found to achieve real-time speed of about 100 frames per second. Embodiments described herein work with x-ray images acquired with and without contrast agents without extra human intervention or extra data.

(34) Embodiments described herein are described with respect to the claimed systems as well as with respect to the claimed methods. Features, advantages or alternative embodiments herein can be assigned to the other claimed objects and vice versa. In other words, claims for the systems can be improved with features described or claimed in the context of the methods. In this case, the functional features of the method are embodied by objective units of the providing system.

(35) Furthermore, certain embodiments described herein are described with respect to methods and

systems utilizing trained machine learning based networks (or models), as well as with respect to methods and systems for training machine learning based networks. Features, advantages or alternative embodiments herein can be assigned to the other claimed objects and vice versa. In other words, claims for methods and systems for training a machine learning based network can be improved with features described or claimed in context of the methods and systems for utilizing a trained machine learning based network, and vice versa.

(36) In particular, the trained machine learning based networks applied in embodiments described herein can be adapted by the methods and systems for training the machine learning based networks. Furthermore, the input data of the trained machine learning based network can comprise advantageous features and embodiments of the training input data, and vice versa. Furthermore, the output data of the trained machine learning based network can comprise advantageous features and embodiments of the output training data, and vice versa.

(37) In general, a trained machine learning based network mimics cognitive functions that humans associate with other human minds. In particular, by training based on training data, the trained machine learning based network is able to adapt to new circumstances and to detect and extrapolate patterns.

(38) In general, parameters of a machine learning based network can be adapted by means of training. In particular, supervised training, semi-supervised training, unsupervised training, reinforcement learning and/or active learning can be used. Furthermore, representation learning (an alternative term is “feature learning”) can be used. In particular, the parameters of the trained machine learning based network can be adapted iteratively by several steps of training.

(39) In particular, a trained machine learning based network can comprise a neural network, a support vector machine, a decision tree, and/or a Bayesian network, and/or the trained machine learning based network can be based on k-means clustering, Q-learning, genetic algorithms, and/or association rules. In particular, a neural network can be a deep neural network, a convolutional neural network, or a convolutional deep neural network. Furthermore, a neural network can be an adversarial network, a deep adversarial network and/or a generative adversarial network.

(40) FIG. 5 shows an embodiment of an artificial neural network **500**, in accordance with one or more embodiments. Alternative terms for “artificial neural network” are “neural network”, “artificial neural net” or “neural net”. Machine learning networks described herein, such as, e.g., the machine learning based feature extractor network utilized at step **104**, the machine learning based correlation network utilized at step **106**, and the machine learning based location predictor network utilized at step **106** of FIG. **1** and feature extraction network **210**, depth wise cross correlation layer **216**, and location predictor network **218** of FIG. **2**, may be implemented using artificial neural network **500**.

(41) The artificial neural network **500** comprises nodes **502-522** and edges **532, 534, . . . , 536**, wherein each edge **532, 534, . . . , 536** is a directed connection from a first node **502-522** to a second node **502-522**. In general, the first node **502-522** and the second node **502-522** are different nodes **502-522**, it is also possible that the first node **502-522** and the second node **502-522** are identical. For example, in FIG. **5**, the edge **532** is a directed connection from the node **502** to the node **506**, and the edge **534** is a directed connection from the node **504** to the node **506**. An edge **532, 534, . . . , 536** from a first node **502-522** to a second node **502-522** is also denoted as “incoming edge” for the second node **502-522** and as “outgoing edge” for the first node **502-522**.

(42) In this embodiment, the nodes **502-522** of the artificial neural network **500** can be arranged in layers **524-530**, wherein the layers can comprise an intrinsic order introduced by the edges **532, 534, . . . , 536** between the nodes **502-522**. In particular, edges **532, 534, . . . , 536** can exist only between neighboring layers of nodes. In the embodiment shown in FIG. **5**, there is an input layer **524** comprising only nodes **502** and **504** without an incoming edge, an output layer **530** comprising only node **522** without outgoing edges, and hidden layers **526, 528** in-between the input layer **524** and the output layer **530**. In general, the number of hidden layers **526, 528** can be chosen

arbitrarily. The number of nodes **502** and **504** within the input layer **524** usually relates to the number of input values of the neural network **500**, and the number of nodes **522** within the output layer **530** usually relates to the number of output values of the neural network **500**.

(43) In particular, a (real) number can be assigned as a value to every node **502-522** of the neural network **500**. Here, $x_{\text{sup.}(n).\text{sub.}i}$ denotes the value of the i -th node **502-522** of the n -th layer **524-530**. The values of the nodes **502-522** of the input layer **524** are equivalent to the input values of the neural network **500**, the value of the node **522** of the output layer **530** is equivalent to the output value of the neural network **500**. Furthermore, each edge **532**, **534**, . . . , **536** can comprise a weight being a real number, in particular, the weight is a real number within the interval $[-1, 1]$ or within the interval $[0, 1]$. Here, $w_{\text{sup.}(m,n).\text{sub.}i,j}$ denotes the weight of the edge between the i -th node **502-522** of the m -th layer **524-530** and the j -th node **502-522** of the n -th layer **524-530**.

Furthermore, the abbreviation $w_{\text{sup.}(n).\text{sub.}i,j}$ is defined for the weight $w_{\text{sup.}(n,n+1).\text{sub.}i,j}$.

(44) In particular, to calculate the output values of the neural network **500**, the input values are propagated through the neural network. In particular, the values of the nodes **502-522** of the $(n+1)$ -th layer **524-530** can be calculated based on the values of the nodes **502-522** of the n -th layer **524-530** by

$$(45) \quad x_j^{(n+1)} = f(\text{Math.}_i \ x_i^{(n)} \cdot \text{Math.} \ w_{i,j}^{(n)}).$$

(46) Herein, the function f is a transfer function (another term is “activation function”). Known transfer functions are step functions, sigmoid function (e.g. the logistic function, the generalized logistic function, the hyperbolic tangent, the Arctangent function, the error function, the smoothstep function) or rectifier functions. The transfer function is mainly used for normalization purposes.

(47) In particular, the values are propagated layer-wise through the neural network, wherein values of the input layer **524** are given by the input of the neural network **500**, wherein values of the first hidden layer **526** can be calculated based on the values of the input layer **524** of the neural network, wherein values of the second hidden layer **528** can be calculated based in the values of the first hidden layer **526**, etc.

(48) In order to set the values $w_{\text{sup.}(m,n).\text{sub.}i,j}$ for the edges, the neural network **500** has to be trained using training data. In particular, training data comprises training input data and training output data (denoted as $t_{\text{sub.}i}$). For a training step, the neural network **500** is applied to the training input data to generate calculated output data. In particular, the training data and the calculated output data comprise a number of values, said number being equal with the number of nodes of the output layer.

(49) In particular, a comparison between the calculated output data and the training data is used to recursively adapt the weights within the neural network **500** (backpropagation algorithm). In particular, the weights are changed according to

$$w'_{\text{sub.}i,j,\text{sup.}(n)} = w_{\text{sub.}i,j,\text{sup.}(n)} - \gamma \cdot \text{Math.} \delta_{\text{sub.}j,\text{sup.}(n)} \cdot \text{Math.} x_{\text{sub.}i,\text{sup.}(n)}$$

wherein γ is a learning rate, and the numbers $\delta_{\text{sup.}(n).\text{sub.}j}$ can be recursively calculated as

$$(50) \quad j^{(n)} = (\text{Math.}_k \ \frac{j^{(n+1)}}{k} \cdot \text{Math.} \ w_{j,k}^{(n+1)}) \cdot \text{Math.} \ f'(\text{Math.}_i \ x_i^{(n)} \cdot \text{Math.} \ w_{i,j}^{(n)})$$

based on $\delta_{\text{sup.}(n+1).\text{sub.}j}$, if the $(n+1)$ -th layer is not the output layer, and

$$(51) \quad j^{(n)} = (x_k^{(n+1)} - t_j^{(n+1)}) \cdot \text{Math.} \ f'(\text{Math.}_i \ x_i^{(n)} \cdot \text{Math.} \ w_{i,j}^{(n)})$$

if the $(n+1)$ -th layer is the output layer **530**, wherein f' is the first derivative of the activation function, and $y_{\text{sup.}(n+1).\text{sub.}j}$ is the comparison training value for the j -th node of the output layer **530**.

(52) FIG. **6** shows a convolutional neural network **600**, in accordance with one or more embodiments. Machine learning networks described herein, such as, e.g., the machine learning based feature extractor network utilized at step **104**, the machine learning based correlation network utilized at step **106**, and the machine learning based location predictor network utilized at step **106** of FIG. **1** and feature extraction network **210**, depth wise cross correlation layer **216**, and location predictor network **218** of FIG. **2**, may be implemented using convolutional neural network

600.

(53) In the embodiment shown in FIG. 6, the convolutional neural network comprises 600 an input layer 602, a convolutional layer 604, and a pooling layer 606. Alternatively, the convolutional neural network 600 can comprise several convolutional layers 604, and several pooling layers 606, as well as other types of layers. The order of the layers can be chosen arbitrarily.

(54) In particular, within a convolutional neural network 600, the nodes 612-616 of one layer 602-606 can be considered to be arranged as a d-dimensional matrix or as a d-dimensional image. In particular, in the two-dimensional case the value of the node 612-616 indexed with i and j in the n-th layer 602-606 can be denoted as $x_{sup.(n).sub.[i,j]}$. However, the arrangement of the nodes 612-616 of one layer 602-606 does not have an effect on the calculations executed within the convolutional neural network 600 as such, since these are given solely by the structure and the weights of the edges.

(55) In particular, a convolutional layer 604 is characterized by the structure and the weights of the incoming edges forming a convolution operation based on a certain number of kernels. In particular, the structure and the weights of the incoming edges are chosen such that the values $x_{sup.(n).sub.k}$ of the nodes 614 of the convolutional layer 604 are calculated as a convolution $x_{sup.(n).sub.k} = K_{sub.k} * x_{sup.(n-1)}$ based on the values $x_{sub.(n-1)}$ of the nodes 612 of the preceding layer 602, where the convolution $*$ is defined in the two-dimensional case as $x_{sub.k.sup.(n)[i,j]} = (K_{sub.k} * x_{sup.(n-1)})[i,j] = \sum_{sub.i'} \sum_{sub.j'} K_{sub.k}[i',j'] \cdot x_{sup.(n-1)}[i-i',j-j']$.

(56) Here the k-th kernel $K_{sub.k}$ is a d-dimensional matrix (in this embodiment a two-dimensional matrix), which is usually small compared to the number of nodes 612-616 (e.g. a 3×3 matrix, or a 5×5 matrix). In particular, this implies that the weights of the incoming edges are not independent, but chosen such that they produce said convolution equation. In particular, for a kernel being a 3×3 matrix, there are only 9 independent weights (each entry of the kernel matrix corresponding to one independent weight), irrespectively of the number of nodes 612-616 in the respective layer 602-606. In particular, for a convolutional layer 604, the number of nodes 614 in the convolutional layer is equivalent to the number of nodes 612 in the preceding layer 602 multiplied with the number of kernels.

(57) If the nodes 612 of the preceding layer 602 are arranged as a d-dimensional matrix, using a plurality of kernels can be interpreted as adding a further dimension (denoted as “depth” dimension), so that the nodes 614 of the convolutional layer 604 are arranged as a (d+1)-dimensional matrix. If the nodes 612 of the preceding layer 602 are already arranged as a (d+1)-dimensional matrix comprising a depth dimension, using a plurality of kernels can be interpreted as expanding along the depth dimension, so that the nodes 614 of the convolutional layer 604 are arranged also as a (d+1)-dimensional matrix, wherein the size of the (d+1)-dimensional matrix with respect to the depth dimension is by a factor of the number of kernels larger than in the preceding layer 602.

(58) The advantage of using convolutional layers 604 is that spatially local correlation of the input data can be exploited by enforcing a local connectivity pattern between nodes of adjacent layers, in particular by each node being connected to only a small region of the nodes of the preceding layer.

(59) In embodiment shown in FIG. 6, the input layer 602 comprises 36 nodes 612, arranged as a two-dimensional 6×6 matrix. The convolutional layer 604 comprises 72 nodes 614, arranged as two two-dimensional 6×6 matrices, each of the two matrices being the result of a convolution of the values of the input layer with a kernel. Equivalently, the nodes 614 of the convolutional layer 604 can be interpreted as arranged as a three-dimensional 6×6×2 matrix, wherein the last dimension is the depth dimension.

(60) A pooling layer 606 can be characterized by the structure and the weights of the incoming edges and the activation function of its nodes 616 forming a pooling operation based on a non-linear pooling function f. For example, in the two dimensional case the values $x_{sup.(n)}$ of the

nodes **616** of the pooling layer **606** can be calculated based on the values $x_{sup.(n-1)}$ of the nodes **614** of the preceding layer **604** as

$$x_{sup.(n)}[i,j] = f(x_{sup.(n-1)}[id_{sub.1}, id_{sub.2}], \dots, x_{sup.(n-1)}[id_{sub.1} + d_{sub.1} - 1, id_{sub.2} + d_{sub.2} - 1])$$

(61) In other words, by using a pooling layer **606**, the number of nodes **614**, **616** can be reduced, by replacing a number $d1.Math.d2$ of neighboring nodes **614** in the preceding layer **604** with a single node **616** being calculated as a function of the values of said number of neighboring nodes in the pooling layer. In particular, the pooling function f can be the max-function, the average or the L2-Norm. In particular, for a pooling layer **606** the weights of the incoming edges are fixed and are not modified by training.

(62) The advantage of using a pooling layer **606** is that the number of nodes **614**, **616** and the number of parameters is reduced. This leads to the amount of computation in the network being reduced and to a control of overfitting.

(63) In the embodiment shown in FIG. **6**, the pooling layer **606** is a max-pooling, replacing four neighboring nodes with only one node, the value being the maximum of the values of the four neighboring nodes. The max-pooling is applied to each d -dimensional matrix of the previous layer; in this embodiment, the max-pooling is applied to each of the two two-dimensional matrices, reducing the number of nodes from **72** to **18**.

(64) A convolutional neural network **600** can also comprise a ReLU (rectified linear units) layer or activation layers with non-linear transfer functions. In particular, the number of nodes and the structure of the nodes contained in a ReLU layer is equivalent to the number of nodes and the structure of the nodes contained in the preceding layer. In particular, the value of each node in the ReLU layer is calculated by applying a rectifying function to the value of the corresponding node of the preceding layer.

(65) The input and output of different convolutional neural network blocks can be wired using summation (residual/dense neural networks), element-wise multiplication (attention) or other differentiable operators. Therefore, the convolutional neural network architecture can be nested rather than being sequential if the whole pipeline is differentiable.

(66) In particular, convolutional neural networks **600** can be trained based on the backpropagation algorithm. For preventing overfitting, methods of regularization can be used, e.g. dropout of nodes **612-616**, stochastic pooling, use of artificial data, weight decay based on the L1 or the L2 norm, or max norm constraints. Different loss functions can be combined for training the same neural network to reflect the joint training objectives. A subset of the neural network parameters can be excluded from optimization to retain the weights pretrained on another datasets.

(67) Systems, apparatuses, and methods described herein may be implemented using digital circuitry, or using one or more computers using well-known computer processors, memory units, storage devices, computer software, and other components. Typically, a computer includes a processor for executing instructions and one or more memories for storing instructions and data. A computer may also include, or be coupled to, one or more mass storage devices, such as one or more magnetic disks, internal hard disks and removable disks, magneto-optical disks, optical disks, etc.

(68) Systems, apparatus, and methods described herein may be implemented using computers operating in a client-server relationship. Typically, in such a system, the client computers are located remotely from the server computer and interact via a network. The client-server relationship may be defined and controlled by computer programs running on the respective client and server computers.

(69) Systems, apparatus, and methods described herein may be implemented within a network-based cloud computing system. In such a network-based cloud computing system, a server or another processor that is connected to a network communicates with one or more client computers via a network. A client computer may communicate with the server via a network browser

application residing and operating on the client computer, for example. A client computer may store data on the server and access the data via the network. A client computer may transmit requests for data, or requests for online services, to the server via the network. The server may perform requested services and provide data to the client computer(s). The server may also transmit data adapted to cause a client computer to perform a specified function, e.g., to perform a calculation, to display specified data on a screen, etc. For example, the server may transmit a request adapted to cause a client computer to perform one or more of the steps or functions of the methods and workflows described herein, including one or more of the steps or functions of FIG. 1 or 2. Certain steps or functions of the methods and workflows described herein, including one or more of the steps or functions of FIG. 1 or 2, may be performed by a server or by another processor in a network-based cloud-computing system. Certain steps or functions of the methods and workflows described herein, including one or more of the steps of FIG. 1 or 2, may be performed by a client computer in a network-based cloud computing system. The steps or functions of the methods and workflows described herein, including one or more of the steps of FIG. 1 or 2, may be performed by a server and/or by a client computer in a network-based cloud computing system, in any combination.

(70) Systems, apparatus, and methods described herein may be implemented using a computer program product tangibly embodied in an information carrier, e.g., in a non-transitory machine-readable storage device, for execution by a programmable processor; and the method and workflow steps described herein, including one or more of the steps or functions of FIG. 1 or 2, may be implemented using one or more computer programs that are executable by such a processor. A computer program is a set of computer program instructions that can be used, directly or indirectly, in a computer to perform a certain activity or bring about a certain result. A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment.

(71) A high-level block diagram of an example computer **702** that may be used to implement systems, apparatus, and methods described herein is depicted in FIG. 7. Computer **702** includes a processor **704** operatively coupled to a data storage device **712** and a memory **710**. Processor **704** controls the overall operation of computer **702** by executing computer program instructions that define such operations. The computer program instructions may be stored in data storage device **712**, or other computer readable medium, and loaded into memory **710** when execution of the computer program instructions is desired. Thus, the method and workflow steps or functions of FIG. 1 or 2 can be defined by the computer program instructions stored in memory **710** and/or data storage device **712** and controlled by processor **704** executing the computer program instructions. For example, the computer program instructions can be implemented as computer executable code programmed by one skilled in the art to perform the method and workflow steps or functions of FIG. 1 or 2. Accordingly, by executing the computer program instructions, the processor **704** executes the method and workflow steps or functions of FIG. 1 or 2. Computer **702** may also include one or more network interfaces **706** for communicating with other devices via a network. Computer **702** may also include one or more input/output devices **708** that enable user interaction with computer **702** (e.g., display, keyboard, mouse, speakers, buttons, etc.).

(72) Processor **704** may include both general and special purpose microprocessors, and may be the sole processor or one of multiple processors of computer **702**. Processor **704** may include one or more central processing units (CPUs), for example. Processor **704**, data storage device **712**, and/or memory **710** may include, be supplemented by, or incorporated in, one or more application-specific integrated circuits (ASICs) and/or one or more field programmable gate arrays (FPGAs).

(73) Data storage device **712** and memory **710** each include a tangible non-transitory computer readable storage medium. Data storage device **712**, and memory **710**, may each include high-speed random access memory, such as dynamic random access memory (DRAM), static random access

memory (SRAM), double data rate synchronous dynamic random access memory (DDR RAM), or other random access solid state memory devices, and may include non-volatile memory, such as one or more magnetic disk storage devices such as internal hard disks and removable disks, magneto-optical disk storage devices, optical disk storage devices, flash memory devices, semiconductor memory devices, such as erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), compact disc read-only memory (CD-ROM), digital versatile disc read-only memory (DVD-ROM) disks, or other non-volatile solid state storage devices.

(74) Input/output devices **708** may include peripherals, such as a printer, scanner, display screen, etc. For example, input/output devices **708** may include a display device such as a cathode ray tube (CRT) or liquid crystal display (LCD) monitor for displaying information to the user, a keyboard, and a pointing device such as a mouse or a trackball by which the user can provide input to computer **702**.

(75) An image acquisition device **714** can be connected to the computer **702** to input image data (e.g., medical images) to the computer **702**. It is possible to implement the image acquisition device **714** and the computer **702** as one device. It is also possible that the image acquisition device **714** and the computer **702** communicate wirelessly through a network. In a possible embodiment, the computer **702** can be located remotely with respect to the image acquisition device **714**.

(76) Any or all of the systems and apparatus discussed herein may be implemented using one or more computers such as computer **702**.

(77) One skilled in the art will recognize that an implementation of an actual computer or computer system may have other structures and may contain other components as well, and that FIG. 7 is a high level representation of some of the components of such a computer for illustrative purposes.

(78) The foregoing Detailed Description is to be understood as being in every respect illustrative and exemplary, but not restrictive, and the scope of the invention disclosed herein is not to be determined from the Detailed Description, but rather from the claims as interpreted according to the full breadth permitted by the patent laws. It is to be understood that the embodiments shown and described herein are only illustrative of the principles of the present invention and that various modifications may be implemented by those skilled in the art without departing from the scope and spirit of the invention. Those skilled in the art could implement various other feature combinations without departing from the scope and spirit of the invention.

Claims

1. A computer-implemented method comprising: receiving first and second input medical images of a patient, the first input medical image comprising an annotation of a location of an object of interest; extracting features from the first and the second input medical images; determining a location of the object of interest in the second input medical image using a machine learning based location predictor network based on the annotation of the location of the object of interest in the first input medical image and the extracted features from the first and the second input medical images, wherein determining a location of the object of interest in the second input medical image using a machine learning based location predictor network based on the annotation of the location of the object of interest in the first input medical image and the extracted features from the first and the second input medical images comprises: correlating the features extracted from the first input medical image with the features extracted from the second input medical image, wherein correlating the features extracted from the first input medical image with the features extracted from the second input medical image comprises: matching features extracted from the first input medical image for regions corresponding to the location of the object of interest with features extracted from the second input medical image, and determining the location of the object of interest in the second input medical image based on the correlated features; and outputting the

location of the object of interest in the second input medical image, wherein the machine learning based location predictor network is trained based on a comparison between 1) locations of a particular object in a sequence of training images determined during a forward tracking of the particular object through the sequence of training images and 2) locations of the particular object determined during a backward tracking of the particular object through the sequence of training images.

2. The computer-implemented method of claim 1, wherein the machine learning based location predictor network determines the locations of the particular object from a first image to a last image of the sequence of training images during the forward tracking and the machine learning based location predictor network determines the locations of the particular object from the last image to the first image in the sequence of training images during the backward tracking.

3. The computer-implemented method of claim 1, wherein the machine learning based location predictor network is further trained based on a comparison between locations of the particular object in annotated training images of the sequence of training images and annotated ground truth locations of the particular object in the annotated training images.

4. The computer-implemented method of claim 1, wherein the machine learning based location predictor network is further trained based on a comparison between a location of the particular object in a first training image of the sequence of training images and an annotated ground truth location of the particular object in the first training image of the sequence of training images.

5. The computer-implemented method of claim 1, wherein a first image and a last image in the sequence of training images are annotated with a ground truth location of the particular object.

6. The computer-implemented method of claim 1, wherein the first and the second input medical images are of a sequence of medical images, the method further comprising: repeating the receiving, the extracting, the determining, and the outputting using the second input medical image as the first input medical image and a next image of the sequence of medical images as the second input medical image.

7. The computer-implemented method of claim 1, wherein the location of the object of interest in the first input medical image is manually annotated by a user or automatically annotated using a machine learning based network.

8. An apparatus comprising: means for receiving first and second input medical images of a patient, the first input medical image comprising an annotation of a location of an object of interest; means for extracting features from the first and the second input medical images; means for determining a location of the object of interest in the second input medical image using a machine learning based location predictor network based on the annotation of the location of the object of interest in the first input medical image and the extracted features from the first and the second input medical images, wherein the means for determining a location of the object of interest in the second input medical image using a machine learning based location predictor network based on the annotation of the location of the object of interest in the first input medical image and the extracted features from the first and the second input medical images comprises: means for correlating the features extracted from the first input medical image with the features extracted from the second input medical image, wherein the means for correlating the features extracted from the first input medical image with the features extracted from the second input medical image comprises: means for matching features extracted from the first input medical image for regions corresponding to the location of the object of interest with features extracted from the second input medical image, and means for determining the location of the object of interest in the second input medical image based on the correlated features; and means for outputting the location of the object of interest in the second input medical image, wherein the machine learning based location predictor network is trained based on a comparison between 1) locations of a particular object in a sequence of training images determined during a forward tracking of the particular object through the sequence of training images and 2) locations of the particular object determined during a backward tracking of

the particular object through the sequence of training images.

9. The apparatus of claim 8, wherein the machine learning based location predictor network determines the locations of the particular object from a first image to a last image of the sequence of training images during the forward tracking and the machine learning based location predictor network determines the locations of the particular object from the last image to the first image in the sequence of training images during the backward tracking.

10. The apparatus of claim 8, wherein the machine learning based location predictor network is further trained based on a comparison between locations of the particular object in annotated training images of the sequence of training images and annotated ground truth locations of the particular object in the annotated training images.

11. The apparatus of claim 8, wherein the machine learning based location predictor network is further trained based on a comparison between a location of the particular object in a first training image of the sequence of training images and an annotated ground truth location of the particular object in the first training image of the sequence of training images.

12. The apparatus of claim 8, wherein a first image and a last image in the sequence of training images are annotated with a ground truth location of the particular object.

13. A non-transitory computer readable medium storing computer program instructions, the computer program instructions when executed by a processor cause the processor to perform operations comprising: receiving first and second input medical images of a patient, the first input medical image comprising an annotation of a location of an object of interest; extracting features from the first and the second input medical images; determining a location of the object of interest in the second input medical image using a machine learning based location predictor network based on the annotation of the location of the object of interest in the first input medical image and the extracted features from the first and the second input medical images, wherein determining a location of the object of interest in the second input medical image using a machine learning based location predictor network based on the annotation of the location of the object of interest in the first input medical image and the extracted features from the first and the second input medical images comprises: correlating the features extracted from the first input medical image with the features extracted from the second input medical image, wherein correlating the features extracted from the first input medical image with the features extracted from the second input medical image comprises: matching features extracted from the first input medical image for regions corresponding to the location of the object of interest with features extracted from the second input medical image, and determining the location of the object of interest in the second input medical image based on the correlated features; and outputting the location of the object of interest in the second input medical image, wherein the machine learning based location predictor network is trained based on a comparison between 1) locations of a particular object in a sequence of training images determined during a forward tracking of the particular object through the sequence of training images and 2) locations of the particular object determined during a backward tracking of the particular object through the sequence of training images.

14. The non-transitory computer readable medium of claim 13, wherein the machine learning based location predictor network determines the locations of the particular object from a first image to a last image of the sequence of training images during the forward tracking and the machine learning based location predictor network determines the locations of the particular object from the last image to the first image in the sequence of training images during the backward tracking.

15. The non-transitory computer readable medium of claim 13, wherein the first and the second input medical images are of a sequence of medical images, the operations further comprising: repeating the receiving, the extracting, the determining, and the outputting using the second input medical image as the first input medical image and a next image of the sequence of medical images as the second input medical image.

16. The non-transitory computer readable medium of claim 13, wherein the location of the object

of interest in the first input medical image is manually annotated by a user or automatically annotated using a machine learning based network.
